Achraf Boutzad

Software Engineer | DevOps Engineer

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Professional Summary

Full-stack Software and DevOps Engineer building real-time production systems, scalable APIs, IoT integrations, and automated delivery pipelines. Skilled in Spring Boot, FastAPI, Next.js, Kafka, MQTT, Docker, Kubernetes, AWS, and CI/CD, with a focus on efficiency, reliability, observability, and measurable factory performance.

Technical Skills

- **Languages:** Java, Python, TypeScript, JavaScript, SQL
- **Backend & APIs:** Spring Boot, FastAPI, REST APIs, GraphQL, gRPC, WebSockets, OpenAPI/Swagger
- **Messaging & Streaming:** Apache Kafka, MQTT, event-driven architecture, asynchronous processing
- **Frontend:** Next.js, React, real-time dashboards, data visualization
- **Databases & Caching:** PostgreSQL, MySQL, MongoDB, Redis, schema design, query optimization
- **DevOps & Containers:** Docker, Docker Compose, Kubernetes, GitHub Actions, CI/CD pipelines
- **Cloud & Infrastructure:** AWS, Terraform (IaC), Nginx, Linux, Git, containerized deployment
- **Observability:** Grafana, Prometheus, Loki, structured logging, health checks, alerting
- **Testing & QA:** JUnit, pytest, Selenium, SonarQube, unit, integration, and API testing
- **Architecture & Practices:** Microservices, data modeling, Agile, Scrum, SDLC, technical documentation
- **Spoken Languages:** Arabic (native), French (B2), English (C1)

Professional Experience

Software Engineer, *Haldorix – MARWA Corporation* Morocco · Jul 2025 - Jul 2026

- Engineered real-time factory dashboards and monitoring workflows that improved factory productivity by up to 80% through bottleneck detection, station visibility, and faster decisions.
- Reduced worker and order waiting time by approximately 30% using live queue, cycle-time, throughput, work-in-progress, and station-efficiency analytics.
- Delivered FastAPI services, WebSocket data flows, workflow automation, Next.js dashboards, Docker, testing, and CI/CD capabilities that supported up to 30% revenue growth and more reliable releases.
- Deployed and operated the platform on Kubernetes with containerized services and automated pipelines, and applied Redis caching to reduce latency on frequently requested dashboard data.

Software Engineering Intern, *Avionav* Tunisia (Remote) · Feb 2025 - Jul 2025

- Developed drone-navigation software using AI-assisted decision logic, A* graph search, waypoint optimization, and obstacle-aware path planning.
- Integrated telemetry and simulation data into testable modules for route generation, trajectory selection, and safer autonomous navigation.

Projects & Analytics Experience

Real-Time Industrial Operations Platform – Next.js, Spring Boot, Kafka, WebSockets, PostgreSQL, Docker

- Built live station and order tracking, throughput and cycle-time analytics, alerts, historical KPIs, automated tests, health checks, logging, and CI/CD deployment.

IoT Monitoring and Predictive Alerting System – FastAPI, Python, MQTT, Kafka, Next.js, Docker

- Built asynchronous telemetry ingestion, device-health monitoring, threshold alerts, event streaming, retries, observability, API documentation, and containerized delivery.

Education

Specialized Master's in Information Systems Engineering 2023 - 2025
Faculty of Sciences Semlalia *Marrakech*

Bachelor's Degree in Mathematics and Computer Science 2018 - 2023
Faculty of Sciences Semlalia *Marrakech*

Certifications

Engineering Practices for Building Quality Software – Coursera | Machine Learning with Python – Coursera